

ZYRO C401 PRO

— Wireless Digital Video Transmission System —

Professional HD FPV Link Solution



1080p60
Video



5.3–5.9 GHz



Betaflight
& INAV



VTX + RBox
+ Camera

ZYRO C401 Product Guide

English Edition with Original Guide Images Preserved



Product Image 1



Product Image 2

C401 Pro User Guide

V2.3

Introduction

The C401 Pro Kit includes a VTX(Wireless Transmitter) and a RBox(Ground Receiver Box), hereinafter referred to as VTX and RBox.

VTX

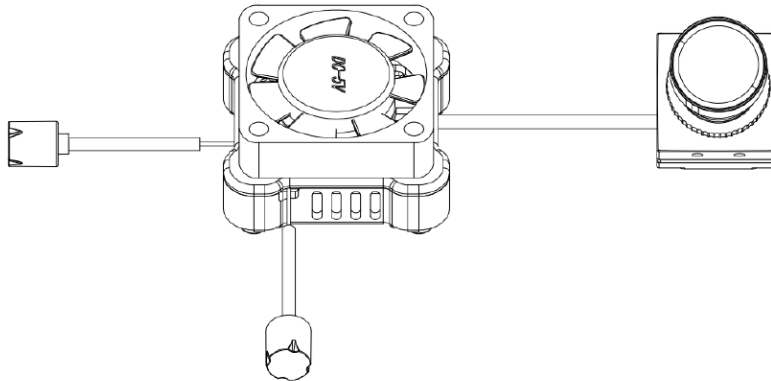
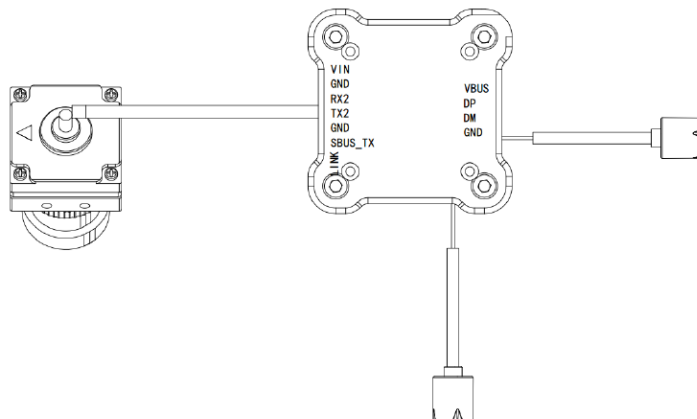


Figure 1.1 VTX appearance



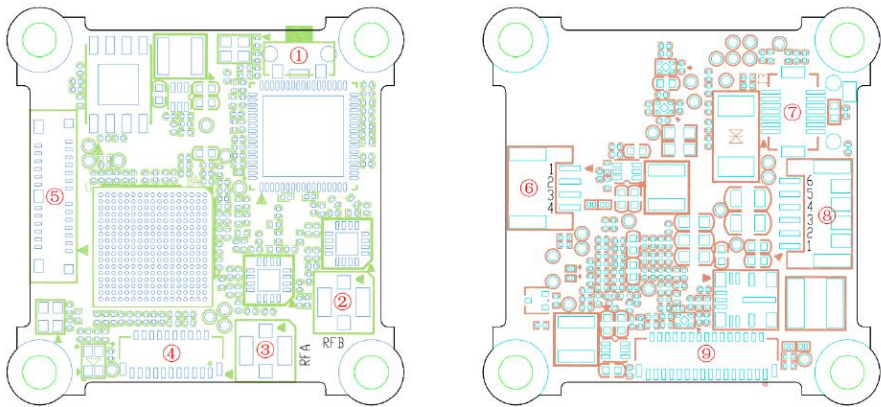


Figure 1.2 VTX Interface

- ① Frequency pairing key
- ② RF portB
- ③ RF portA
- ④ Debug interface
- ⑤ Camera interface
- ⑥ USB_CON

Signal Name	Description
GND	Ground
USB_DN	USB data negative signal
USB_DP	USB data positive signal
USB_VBUS	USB VBUS signal

- ⑦ BTB connector used to mount SD card board
- ⑧ PWR_CON

Signal Name	Description
PWR_IN	VTX input power 8 ~ 30V
GND	Ground

Notes

1. Before powering on the VTX and RBox, ensure the antennas are connected correctly. Otherwise, the devices may be damaged.

Frequency Pairing

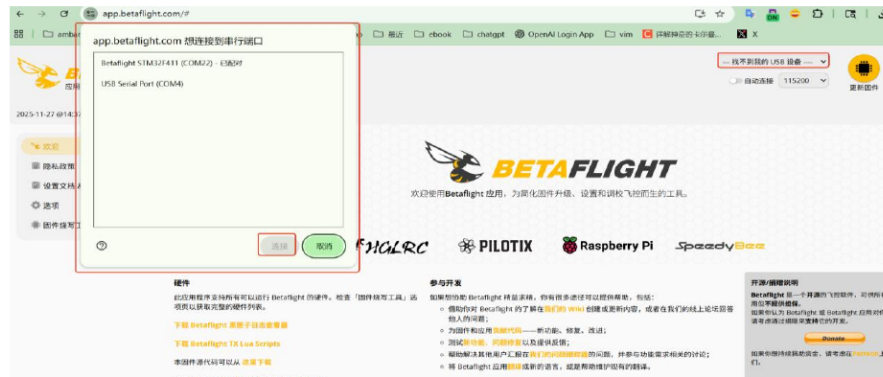
1. Connect the power supplies of the VTX and RBox.
2. Briefly press the frequency pairing buttons on the VTX and RBox respectively. After entering the frequency pairing state, the dual-color LEDs of the VTX and RBox will flash rapidly.
3. When the wireless connection is successful, the green LEDs of both the VTX and RBox will stay on, indicating a normal wireless connection.

Firmware Upgrade

1. Copy the upgrade files of the VTX and RBox to the root directory of the SD card. It is recommended that the SD card has been formatted inside the RBox so that it can be correctly recognized by the RBox. Note that the firmware names must be "A1_SKY_v1.x.x_2026xxx.img" and "A1_GND_v1.x.x_2026xxx.img" respectively.
2. Keep the VTX and RBox in a normal wireless connection state, insert the SD card into the RBox, and ensure the SD card is correctly recognized (the icon in the lower right corner will turn blue and display the capacity).
3. Select [Settings] -> [System] -> [System Upgrade] in the menu, and click Confirm in the pop-up menu.
4. Wait for the upgrade to complete. After the upgrade is finished, a menu pop-up will prompt the user to restart the VTX and RBox. The new firmware will take effect after restart, and the normal wireless connection will be maintained without the need for re-pairing.
5. The firmware of the VTX and RBox can be upgraded separately. That is, whichever firmware is placed in the SD card, the system will upgrade the corresponding device. Device without the corresponding firmware will not be upgraded.
6. After the upgrade, you can view the current firmware version information through the menu [Settings] -> [System] -> [About].

Betaflight Output MSP OSD Setting

1. Solder the white and gray wires of the 4 pin cable to the flight controller.
2. Configure flight controller by <https://app.betaflight.com>. If your google chrome browser cannot find the Betaflight controller, you can follow below steps.

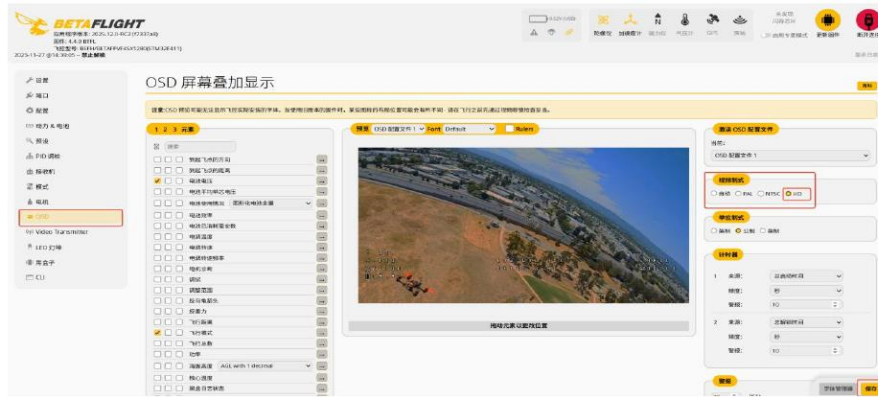


3. Connect the flight controller by the Betaflight Configurator, choose any serial port as the MSP display port, taking UART 2 as an example.



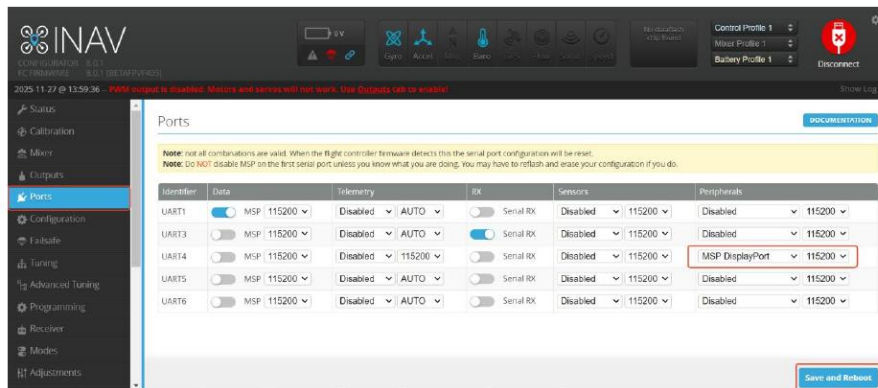
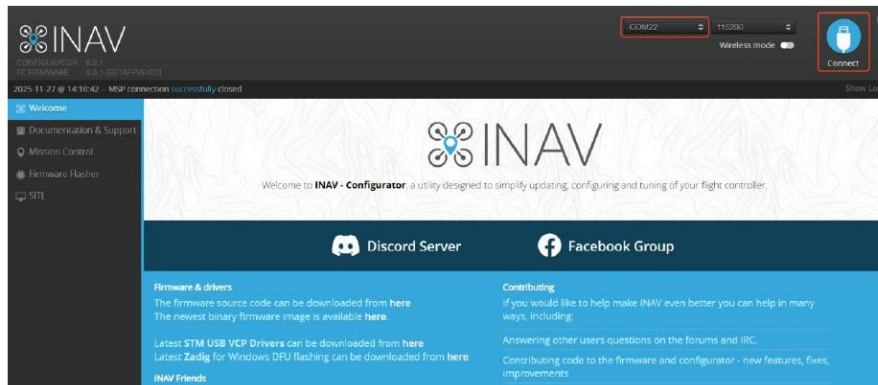
4. Currently, our UI OSD font only supports “HD” video format. Save and restart the flight controller, and MSP OSD output is setting done.

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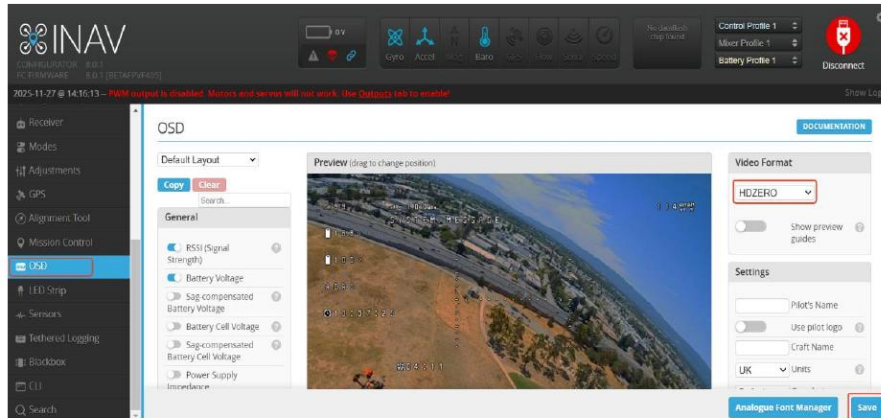
INAV Output MSP OSD Setting

1. Solder the cable to the INAV flight controller.
2. Connect the flight controller by the INAV Configurator, choose any serial port as the MSP display port, taking UART 4 as an example.



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- Currently, our UI OSD font only supports “HDZERO” video format. Save and restart the flight controller, and MSP OSD output is setting done.



LED Status Indicators

Status	Description
Unpaired State	The red LEDs of both the VTX and RBox are always on
Pairing in Progress	The dual-color LEDs of both the VTX and RBox flash rapidly
Pairing Successful	The green LEDs of both the VTX and RBox are always on

VTX Specifications

Model	C401 Pro
Communication Frequency	5.3 ~ 5.9GHz
Transmit Power(EIRP)	FCC: <30dBm; CE: <14dBm; SRRC: <20dBm; MIC: <25dBm
Interfaces	JST1.0*4 (Power Cable)
Mounting Hole Spacing	25.5mm x 25.5mm
Dimensions	31.8mm x 31.8mm x 11.3mm
Video Resolution	1080p60fps/720p60fps
Weight	21.8g

Operating Environment Temperature	-20 ~ 60℃
Number of Frequency	19
Power Input	8 ~ 30V
Supported Flight Control System	Betaflight, INAV
OSD	Canvas Mode
Antennas	2 * IPEX-1

RBox Specification

Model	C401 Pro
Communication Frequency	5.3 ~ 5.8GHz
SD Card Slot	Supports up to 1TB capacity
Type-C Interface	USB-A to Type-C Cable to PC for software upgrade
HDMI Interface	Video output
Power Input	8 ~ 30V
Antennas	2 * IPEX-1

Camera Specifications

Image Sensor	1/1.8 inch, 2MP
Resolution	1080p60fps/720p60fps
Ratio	16:9/4:3
FOV	160°
Aperture	F2.2
Shutter	Rolling Shutter
Weight	13.47g
Dimensions	19mmx19mmx24mm

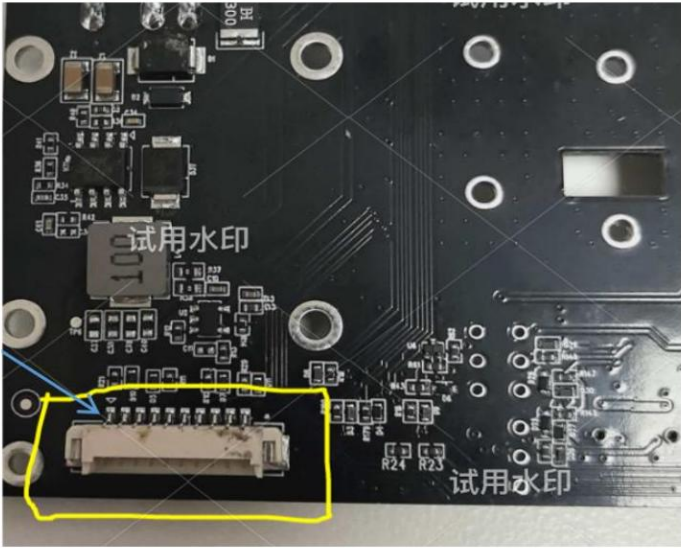
Minimum Illumination	0.0001Lux
Coaxial Cable Length	140mm

Antenna Specifications

Polarization	LHCP
Bandwidth	5.1 ~ 5.9GHz
Average Gain	2dBi
Input Impedance	50 Ω
VSWR	<1.5
Interface	IPEX-1

RBox Button Interface

The position of the button interface: on the back of the button, the yellow frame area, the arrow indicating pin 1.



Number	Button	Function
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1	K1	Pair
2	K2	Up
3	K4	Left
4	K3	Back
5	K6	Record
6	K5	Down
7	K8	Right
8	K7	Enter
9	GND	Ground
10	-	-